

Structural Primers as Procedural Shortcuts in Language-Model Graph Reasoning

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Abstract

Graph descriptors written before an adjacency list may supply the answer to a query, redirect its solution procedure, or erase distinctions in a continuous feature. We synthesize a graph-blind primer control, response-pattern classifications, and a paired degree-neighborhood invariant test over 84,000 unique, target-verified Qwen3 generations on 40-node graphs. A degree primer increases plain Qwen3-4B edge-count success from 2.00% to 29.25%, but decreases its jointly correct degree and neighborhood answers from 85.75% to 76.50%. On degree queries, enumeration gives way to retrieval-style responses: at density $p = .50$, accuracy falls from 99% to 87%. Rounding random-walk return features to two decimals leaves only 3.30 distinct feature pairs among 40 nodes per graph at $p = .50$, versus 40.00 at six decimals. These are measured failures of reliable execution and feature resolution, not evidence that a single primer generally improves graph understanding.

1 Introduction

Graph machine learning uses degrees, local motifs, and random-walk statistics to expose relations that a node’s text alone cannot express. When the same descriptors are serialized as sentences before an adjacency list, a language model (LLM) can instead copy a stated answer, follow a different procedure, or misread a rounded number. Accuracy on one question does not show whether another answer about the same graph obeys a necessary identity. We ask: *How do textual structural primers alter an LLM’s procedural execution of graph tasks, and what are the expressivity limits of these primers in preserving structural invariants?*

The distinction matters when evaluations equate access to graph text with algorithmic use of the graph. We join two diagnostics on the same saved GraphTalk corpus: an edge-list-blind solver that identifies answer-carrying primers, and a paired

consistency measure that checks whether predicted degree equals predicted neighborhood size. Independently reconstructed graph targets, capped exact-match scoring, and response classifications connect these diagnostics to observed outputs. We study 84,000 *unique* generations, four model/mode arms, seven prompt conditions, and edge probabilities $p = .10-.85$. The central result is a trade-off: information that improves global counting can damage locally related answers. A separate precision audit identifies an information bottleneck before the LLM even begins the query.

2 Prior Work and the Gap

GraphQA shows that text encodings and graph tasks change LLM accuracy (Fatemi et al., 2024); NLGraph and GraphArena test further graph computations (Wang et al., 2023; Tang et al., 2025). These evaluations establish task sensitivity, but aggregate success alone does not distinguish answer extraction from adjacency-based execution or test whether two outputs obey the same graph axiom. Huang et al. (2024) already distinguish label-relevant text from structural use in node classification, so our claim is narrower: a joint diagnostic of answer availability, output procedure, cross-query correctness, and serialization resolution on matched graph queries.

Graph neural architectures use local and positional descriptors to increase accessible structural information (Dwivedi et al., 2022; Rampásek et al., 2022). Here, degree, clustering, and random-walk structural encoding (RWSE) are *textual* side channels to a frozen LLM; their presence does not confer GNN expressivity. GraphToken instead learns a graph-to-LLM interface (Perozzi et al., 2024). Shortcut learning names predictive reliance on an easier cue (Geirhos et al., 2020), but surface response patterns cannot establish a model’s latent algorithm. Context length can also affect accuracy (Levy et al., 2024; Shi et al., 2023); a node-list

filler supplies a partial control, not an exact token-matched experiment.

The representation level matters. A GNN receives numeric descriptors together with a topology over which it propagates; the LLM sees rounded sentences followed by a separate adjacency list. Identical printed RWSE vectors remove a distinction from that descriptor channel, even though the adjacency sentences may still distinguish their vertices. Conversely, a printed degree can leak an exact degree-query target. Task averages alone cannot tell whether either case occurred. We measure answer availability, observable response style, related-output validity, and descriptor resolution on the same graph series.

3 Data and Method

Matched graph queries. The main sweep uses 100 independently seeded, undirected $G(40, p)$ graphs at each $p \in \{.10, .20, .35, .50\}$; every graph is reused across Qwen3-1.7B and Qwen3-4B (Yang et al., 2025), each in plain and native thinking modes. Six tasks ask node count, edge count, a target node’s degree, its neighbor set, existence of one edge, and existence of a cycle. The three local/edge questions use recorded query endpoints; degree and neighbors share a target node. Conditions are no primer, filler enumerating the nodes, number of components, per-node clustering coefficient, per-node RWSE, per-node degree, and ALL (degree, clustering, RWSE). The graph follows the primer as an incident adjacency list. RWSE prints $((D^{-1}A)_{vv}^2, (D^{-1}A)_{vv}^3)$ to two decimals after internal six-decimal rounding (zero rows for isolates); clustering is also printed to two decimals. Component count gives no component membership.

This yields $4 \times 4 \times 100 \times 6 \times 7 = 67,200$ main responses. The extension has $p \in \{.65, .75, .85\}$ and *only* degree and edge-existence tasks, giving $4 \times 3 \times 100 \times 2 \times 7 = 16,800$ more. Audit of the repository’s¹ 84,002 raw records removes two identical duplicates, yielding 84,000 unique records; independent seeded reconstruction agrees with all saved gold targets. The 40-node and cycle gold labels are constant in all 700 graphs, so these two tasks cannot diagnose structural reasoning and are omitted from comparative results.

For traceability, the audit reads 172

`densfull140[hi].shard*.jsonl` files and retains each response’s source filename and line number. It keys deduplication by arm, graph/query instance, and primer condition; two repeated records are identical. Graph reconstruction uses seed $20260906 + 10^6(1 + \text{round}(1000p)) + 40000 + i$ at density p and graph index $i = 0, \dots, 99$. Successive draws from Python’s `random.Random` determine edges over unordered vertex pairs; an independently initialized RNG chooses the query target. This audits saved outputs; it creates no new model responses.

Success and inference. The primary indicator is $S_{ict} = 1[\hat{y}_{ict} = y_{it}]1[\text{uncapped}]$ for graph i , condition c , and task t . Degree and count require exact integers; neighbors require exact set equality. Invalid or unparsed responses fail. The saved main sweep uses greedy generation and an 8,192-new-token cap. In the extension, plain arms use 2,048 and thinking arms 8,192. No model is rerun. We contrast paired outputs on the same graphs; 95% intervals are density-stratified paired bootstrap intervals (4,000 resamples; seed 1947). Exact McNemar tests receive Benjamini–Hochberg correction over six primer-versus-none comparisons separately for each model/task or diagnostic family (McNemar, 1947; Benjamini and Hochberg, 1995). This is exploratory, not experiment-wide error control. Completion-conditioned accuracies are reported separately and never substituted for S .

Answer-carrying control. The graph-blind solver parses only the *rendered* primer and available counts, never the adjacency list; it is denied the count whose value a question requests. For degree, the degree sentence directly supplies d_v ; for edge count, the degree list supplies $|E| = \frac{1}{2} \sum_v d_v$. These are answer-carrying even though the LLM may fail to extract or sum them. Clustering, RWSE, and component count do not directly state either target. This control diagnoses leakage of target information, not a learned model’s competence. The supplied no-primer and filler prompts contain the node set; filler is consequently only a verbosity comparison.

On the reconstructed targets, the degree primer permits exact graph-blind recovery of all 700 degree-query targets and all 400 main-sweep edge-count targets. ALL contains those same recoverable values. These are analytical control outcomes for

¹<https://github.com/GalBrk/GraphTalk>, inspected at commit d9b2190f3ad0e40c0392159839a9c6167132704e.

the 700 existing graphs, not extra LLM generations. The baseline with no primer contains neither a per-node degree list nor its sufficient sum. The control establishes that a success gain *can* arise from copied or summed side-channel values; it does not show that the LLM actually performs either operation.

Response-pattern coding. For a terminated degree response, enumeration requires at least five neighbors written one per line. Retrieval style means the response answers without restating the queried node’s neighbor list; remaining responses form an uncoded residual. These are observable text patterns, not validated internal algorithms. Their reported percentages use terminated outputs in the relevant density/condition cell; exact-match success retains all 100 outputs, including capped failures.

Related-output diagnostic. For the same graph and target node, let $\hat{d}_i$ and $\hat{N}_i$ be predictions in two separate prompts. Set $C_i = 1$ if both complete and parse, all predicted node IDs lie in $0, \dots, 39$, and $\hat{d}_i = |\hat{N}_i|$. Set $J_i = S_{i,\text{degree}} S_{i,\text{neighbors}}$. Then $J_i \leq C_i$, and $C - J$ counts consistent yet jointly incorrect pairs. C is a necessary cardinality identity; satisfying it does not prove that the predicted neighbors are actual edges. Each cell has 400 paired graphs in the main sweep.

4 Results

Direct information redirects execution. Table 1 shows degree information helping plain 4B global edge count by 27.25 points, from 2.00% to 29.25% (95% CI [22.75, 31.75]; 114 failed instances corrected, five successes lost). Yet the same model’s degree accuracy falls by 6.50 points, from 99.25% to 92.75%. The 1.7B plain arm gains only 3.00 points on edge count ($q = .101$); its thinking arm loses 3.25. Thus supplying a mathematically sufficient statistic does not guarantee execution of its forty-term sum. Neither ALL nor filler universally helps: plain-4B ALL reaches just 5.75% on edge count, against 29.25% with degree alone. In 4B thinking, clustering and components each reach 44.75% edge-count success versus 20.00% without a primer, although neither states the edge count; that 24.75-point improvement cannot be interpreted as simple copying.

Observed response styles locate the change more sharply. For plain 4B degree queries without a

Arm	Task	None	Degree	All
1.7B	Degree	60.50	68.00	70.75
	Neighbors	72.50	76.00	74.50
	Edge exists	70.50	74.50	82.00
	Edge count	1.25	4.25	2.00
1.7B-T	Degree	76.25	85.00	88.50
	Neighbors	78.75	85.00	85.00
	Edge exists	99.00	96.50	97.25
	Edge count	15.75	12.50	20.75
4B	Degree	99.25	92.75	91.25
	Neighbors	86.00	83.00	83.25
	Edge exists	90.50	85.75	80.50
	Edge count	2.00	29.25	5.75
4B-T	Degree	97.00	99.50	97.50
	Neighbors	95.50	96.00	94.25
	Edge exists	99.75	99.75	98.75
	Edge count	20.00	28.50	21.75

Table 1: Exact, uncapped success (%) on the main sweep, $n = 400$ per cell. T: native thinking. Degree and All denote primer conditions, not estimates conditioned on termination.

p	None	Degree	Enum. N	Enum. D	Retr. D
.10	100	94	1	2	44
.20	99	98	34	16	56
.35	99	92	95	6	74
.50	99	87	100	8	62
.65	82	76	94	2	71
.75	50	57	85	1	64
.85	33	60	44	0	69

Table 2: Plain-4B degree-query accuracy and response styles (%; 100 graphs per density). Enum.: lists at least five neighbors on separate lines; Retr.: answers without restating the queried neighbor list. N/D: no primer/degree primer. Style shares are conditional on termination; accuracy scores all saved outputs.

primer, explicit multi-line neighbor enumeration occurs in 95% and 100% of responses at $p = .35$ and $.50$, with 99% exact accuracy in either stratum. Under a degree primer, enumeration falls to 6% and 8%, and retrieval-style answers (answer without restating the queried node’s neighbor list) occur in 74% and 62%. At $.50$ the accuracy drops from 99% to 87% (one newly correct, 13 newly wrong). At $.85$, the no-primer accuracy is 33%, compared with 60% with degree; 69% of primer responses are retrieval-style. The observed gains and losses follow the relative accuracy of these response styles, not a uniform informational benefit. The style classifier detects text patterns, not the hidden computation, and the $.85$ result also has a different plain-arm cap and longer adjacency lists.

The style shift does not generalize to every arm. Under degree, 1.7B thinking still enumerates neigh-

Arm	None		Degree	
	C	J	C	J
1.7B	62.25	54.00	73.00	62.50
1.7B-T	81.00	71.00	81.00	76.50
4B	87.00	85.75	81.00	76.50
4B-T	94.00	93.75	96.00	95.75

Table 3: Degree–neighborhood cardinality agreement C and joint correctness J (% of 400 paired graphs per cell). $C - J$ is consistent but wrong.

bors in 76–98% of terminated degree responses from $p = .35$ to $.85$, and reports a mismatch between its own count and the primer in 40–59% of those density cells. Over the four main densities, its unconditional degree success increases from 76.25% to 85.00% (+8.75 points), while plain 4B declines from 99.25% to 92.75% (−6.50 points). Thus the same answer-carrying string accompanies different observable procedures and opposite aggregate effects.

Correctness is stricter than agreement. Table 3 exposes a negative local result masked by edge-count improvement. For plain 4B, degree reduces C from 87.00% to 81.00%, and J from 85.75% to 76.50%: a −9.25-point paired change in joint correctness (95% CI $[-13.75, -4.75]$; 28 pairs fixed, 65 broken). The contrast survives the specified multiple-test adjustment. For plain 1.7B the same primer raises J from 54.00% to 62.50%, yet $C - J$ grows from 8.25 to 10.50 points: more pairs obey cardinality while at least one output is wrong. Under plain-4B components, J reaches 95.00% (a +9.25-point change), despite this global count not giving either local answer. Therefore a primer’s utility is conditional on task and mode; even internally compatible answers can be wrong about the graph.

Completion and label prevalence qualify gains. In no-primer edge counting, 4B thinking completes only 20.50% of outputs; 97.56% of those completions are exact, leaving 20.00% unconditional success. The 1.7B thinking figures are 18.25% completion, 86.30% conditional accuracy, and 15.75% success. The cap-sensitive result cannot establish what a larger budget would do. In plain 1.7B edge-existence queries, raw success rises from 54% at $p = .50$ to 86% at $.85$, but balanced accuracy is 55.77% and 53.33%, respectively. The model says “yes” on 94% and 99% of queries; at $.85$, 85 of 100 queried pairs are edges, so an always-yes pre-

dictor already obtains 85%. This apparent density recovery does not demonstrate better rejection of nonedges.

Edge-existence accuracy is similarly sensitive to false positives. Across seven densities, plain 1.7B’s ALL primer lowers false positive edge predictions from 51% to 33%, a −18.6-point change (95% interval $[-24.0, -13.1]$); filler raises them to 69%, a +18.3-point change ($[13.9, 22.7]$). Clustering in plain 4B lowers this error rate from 16% to 8% and raises completed-pair edge-existence accuracy by 4.6 points ($[2.3, 6.9]$). These differences constrain interpretation of raw accuracy because prevalence of actual edges changes sharply with p .

Text discretizes continuous structure. In reconstructed graphs, the mean number of distinct printed two-decimal RWSE pairs among 40 nodes is 30.14 at $p = .10$, 3.30 at $.50$, and 2.00 at $.85$. At $.50$ the modal printed pair describes 62.68% of vertices on average. Reprinting the *same computed features* to six decimals yields 40.00 distinct pairs at $.50$ and 39.99 at $.85$. Thus a solver observing only the two-decimal RWSE field cannot distinguish the collided vertices *by that field*. This is a feature-interface resolution bound, not a proof about graph isomorphism, downstream GNN expressivity, or the causal source of observed RWSE task accuracy; no higher-precision LLM arm was run.

Because adjacency sentences and IDs can still distinguish vertices sharing one printed RWSE pair, the measured collision is confined to that feature field. Extra printed digits also lengthen the input, so the six-decimal counterfactual is a property of feature strings, not a measured LLM accuracy intervention.

5 Discussion and Limitations

The paired edge-count and local-query outcomes support a specific interpretation: printed structural information can induce a cheaper observable response style whose reliability differs from the adjacency procedure it replaces. Direct degree leakage plus the +27.25-point global count effect cannot certify learned counting; the simultaneous −9.25-point drop in local joint correctness rules out a general accuracy benefit in plain 4B. For 4B thinking, non-answer-carrying components and clustering each yield +24.75 points on edge count, so prompt framing or completion changes remain plausible. The response classifier only reads generated text

and cannot identify internal algorithms.

Our claims apply to one incident-list ordering, 40-node Erdős-Rényi graphs, Qwen3-1.7B/4B, and the saved deterministic generations. Density changes list length and task prevalence; in the extension it also changes the plain-arm cap from 8,192 to 2,048. No vertex relabeling, different graph family, equal-compute cap, precision intervention, node-classification training, or held-out link prediction is measured. The original answer extractor can misparse some neighborhood lists; a targeted manual sensitivity audit shifts C or J by at most 1.25 points in any inspected cell and leaves the plain-4B cells unchanged, but is not an exhaustive parser validation. Bootstrap intervals describe variation across sampled graphs, not model-training variability. A decisive follow-up would cross feature precision with matched token budgets, balanced edge queries, and node permutations while retaining the paired invariant test.

6 Conclusion

The 84,000-response audit separates answer-carrying text, observed response style, cross-query validity, and feature resolution. Degree text raises plain-4B edge-count success from 2.00% to 29.25%, but lowers its joint local correctness from 85.75% to 76.50%. At $p = .50$, two-decimal RWSE preserves only 3.30 distinct pairs across 40 vertices on average. Together these results support a bounded thesis: primers can redirect observable response styles toward brittle shortcuts without ensuring preservation of the graph relations queried elsewhere.

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